SE Lab 5

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Class: 5H

Setup

```
(venv) PS C:\Users\vk360\Desktop\College\Lab Work\SE\Lab 5 - Oct 14th\L5_14_recruitment_tracker-main> python recruitment_tracker.py
['Nina']
2 in interview
```

```
(venv) PS C:\Users\vk360\Desktop\College\Lab Work\SE\Lab 5 - Oct 14th\L5_14_recruitment_tracker-main> pylint recruitment_tracker.py > pylint_report_before.txt
(venv) PS C:\Users\vk360\Desktop\College\Lab Work\SE\Lab 5 - Oct 14th\L5_14_recruitment_tracker-main> bandit -r recruitment_tracker.py > bandit_report.txt
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.13.7
(venv) PS C:\Users\vk360\Desktop\College\Lab Work\SE\Lab 5 - Oct 14th\L5_14_recruitment_tracker-main> flake8 recruitment_tracker.py > flake8_report_before.txt
(venv) PS C:\Users\vk360\Desktop\College\Lab Work\SE\Lab 5 - Oct 14th\L5_14_recruitment_tracker-main>
```

Bandit (before)

```
Code scanned:
    Total lines of code: 38
    Total lines skipped (#nosec): 0
    Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
    Total issues (by severity):
        Undefined: 0
        Low: 1
        Medium: 2
        High: 0
    Total issues (by confidence):
        Undefined: 0
        Low: 0
        Medium: 1
        High: 2
Files skipped (0):
```

Flake8 (before)

```
1  recruitment_tracker.py:4:80: E501 line too long (124 > 79 characters)
2  recruitment_tracker.py:8:80: E501 line too long (140 > 79 characters)
3  recruitment_tracker.py:13:80: E501 line too long (106 > 79 characters)
4  recruitment_tracker.py:15:5: F841 local variable 'total_seen' is assigned to but never used
5  recruitment_tracker.py:15:27: W291 trailing whitespace
6  recruitment_tracker.py:26:80: E501 line too long (93 > 79 characters)
7  recruitment_tracker.py:31:1: E302 expected 2 blank lines, found 1
8  recruitment_tracker.py:39:80: E501 line too long (100 > 79 characters)
```

Pylint (before)

pylint_report.txt

before > pylint_report.txt

```
************* Module recruitment_tracker
recruitment_tracker.py:4:0: C0301: Line too long (124/100) (line-too-long)
recruitment_tracker.py:8:0: C0301: Line too long (140/100) (line-too-long)
recruitment_tracker.py:13:0: C0301: Line too long (106/100) (line-too-long)
recruitment_tracker.py:15:26: C0303: Trailing whitespace (trailing-whitespace)
recruitment_tracker.py:1:0: C0114: Missing module docstring (missing-module-docstring)
recruitment_tracker.py:7:0: C0116: Missing function or method docstring (missing-function-docstring)
recruitment_tracker.py:7:0: C0103: Function name "addApplicant" doesn't conform to snake_case naming style (invalid-
recruitment_tracker.py:12:0: C0116: Missing function or method docstring (missing-function-docstring)
recruitment_tracker.py:14:12: W0123: Use of eval (eval-used)
recruitment_tracker.py:12:0: R1710: Either all return statements in a function should return an expression, or none
recruitment_tracker.py:15:4: W0612: Unused variable 'total_seen' (unused-variable)
recruitment_tracker.py:24:0: C0116: Missing function or method docstring (missing-function-docstring)
recruitment_tracker.py:24:0: C0103: Function name "updateStatus" doesn't conform to snake_case naming style (invalic
recruitment_tracker.py:28:11: W0718: Catching too general exception Exception (broad-exception-caught)
recruitment_tracker.py:31:0: C0116: Missing function or method docstring (missing-function-docstring)
recruitment_tracker.py:31:0: C0103: Function name "exportReport" doesn't conform to snake_case naming style (invalic
recruitment_tracker.py:32:8: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
recruitment_tracker.py:32:8: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
recruitment_tracker.py:38:0: C0116: Missing function or method docstring (missing-function-docstring)
recruitment_tracker.py:38:0: C0103: Function name "countByStatus" doesn't conform to snake_case naming style (invali
recruitment_tracker.py:42:0: C0116: Missing function or method docstring (missing-function-docstring)

-----------------------------------
Your code has been rated at 4.47/10
```

recruitment_tracker.py (before)

```python
def addApplicant(name, role, expected_salary, status="applied"):
    conn.execute(f"INSERT INTO applicants (name, role, expected_salary, status) VALUES ('{name}', '{role}', {expecte
    conn.commit()


def shortlist(role, salary_rule):
    rows = conn.execute("SELECT name, expected_salary FROM applicants WHERE role = ?", (role,)).fetchall()
    limit = eval(salary_rule)
```

Bandit (after)

bandit_report_after.txt U

bandit_report_after.txt

```
Run started:2026-10-14 07:09:24.317615+00:00

Test results:
    No issues identified.

Code scanned:
    Total lines of code: 37
    Total lines skipped (#nosec): 0
    Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
    Total issues (by severity):
        Undefined: 0
        Low: 0
        Medium: 0
        High: 0
    Total issues (by confidence):
        Undefined: 0
        Low: 0
        Medium: 0
        High: 0
Files skipped (0):
```

Pylint (after)

pylint_report_after.txt U

pylint_report_after.txt

```

--------------------------------------------------------------------
Your code has been rated at 10.00/10 (previous run: 10.00/10, +0.00)

```

Flake8 (after)

flake8_report_after.txt U

flake8_report_after.txt

```

```

Known Issue Table

Issue	Type	Line(s)	Description	Fix Approach
SQL injection (Bandit B608, Medium)	Security	8	The INSERT is built with an f-string from user values. O'Brien breaks it, and a crafted value can run arbitrary SQL.	Parameterised query with ? placeholders and a tuple of values
Use of eval (Bandit B307 Medium, Pylint W0123)	Security	14	eval(salary_rule) executes any string it is given, which allows arbitrary code execution.	Remove eval; shortlist() takes a numeric salary_limit. Bandit's suggested ast.literal_eval fails on "500000 + 150000"; I tested it and it raises ValueError.
try/except/pass, broad exception (Bandit B110 Low, Pylint W0718)	Security / Bug	28–29	Every error is swallowed, so a failed update fails silently.	Catch only sqlite3.Error and report it
Inconsistent return (Pylint R1710)	Bug	12, 20–21	Returns a list when there are matches, otherwise None.	Always return a list (empty if no match)
No with / no encoding (Pylint	Bug	32–35	The file handle leaks if a write fails, and the encoding depends on the platform.	with open(path, "w", encoding="utf-8")

R1732, W1514)				
Unused variable (Pylint W0612, Flake8 F841)	Code quality	15	total_seen is computed but never used.	Delete the line
Non-snake_case names (Pylint C0103)	Style	7, 24, 31, 38	addApplicant, updateStatus, exportReport, countByStatus	Rename to snake_case and update the calls
Missing docstrings (Pylint C0114, C0116)	Documentation	1, 7, 12, 24, 31, 38, 42	The module and all six functions have none.	Add docstrings
Line too long (Flake8 E501, Pylint C0301)	Style	4, 8, 13, 26, 39	Lines are 93–140 characters long.	Wrap strings and calls inside parentheses
Trailing whitespace (Flake8 W291, Pylint C0303)	Style	15	Spaces at the end of the line	Removed along with the unused variable
Expected 2 blank lines	Style	31	Only one blank line before def exportReport	Add a blank line

(Flake8 E302)				

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